

Giving Molecule Design a Sense of Space — and a Sense of Curiosity

ICLR 2022: DGAPN pairs spatial graph attention with a random-network-distillation curiosity bonus to design stronger, more synthesizable SARS-CoV-2 inhibitors

Designing a molecule that binds tightly to a viral protein — and that a chemist could actually make — is, at heart, a search problem over an almost unimaginably large space. The number of drug-like molecules is astronomical, and each one is not a vector of numbers but a graph of atoms joined by bonds. Searching that discrete, physically constrained space efficiently is the obstacle every automated drug-discovery method runs into.

Generative and reinforcement-learning approaches to "growing" molecules have multiplied in recent years, but two things kept holding them back. First, most graph neural networks look only at atom features and connectivity — they ignore bond attributes, and they ignore the molecule's three-dimensional shape, even though whether a candidate fits a receptor pocket depends precisely on its geometry and chemical complementarity. Second, many methods build a molecule one atom and one bond at a time, which yields long, unstable trajectories and molecules that are hard to synthesize.

This ICLR 2022 paper introduces DGAPN — the Distilled Graph Attention Policy Network — to close both gaps at once. It is built from three pieces: spatial Graph Attention (sGAT), a GNN that encodes 3D structure alongside chemistry; the Graph Attention Policy Network (GAPN), an attentional policy over a fragment-level action space; and a curiosity-driven exploration bonus implemented with Random Network Distillation (RND). On the task of designing inhibitors for the SARS-CoV-2 NSP15 protein, DGAPN beats all five state-of-the-art baselines on binding affinity while producing molecules that are easier to make.

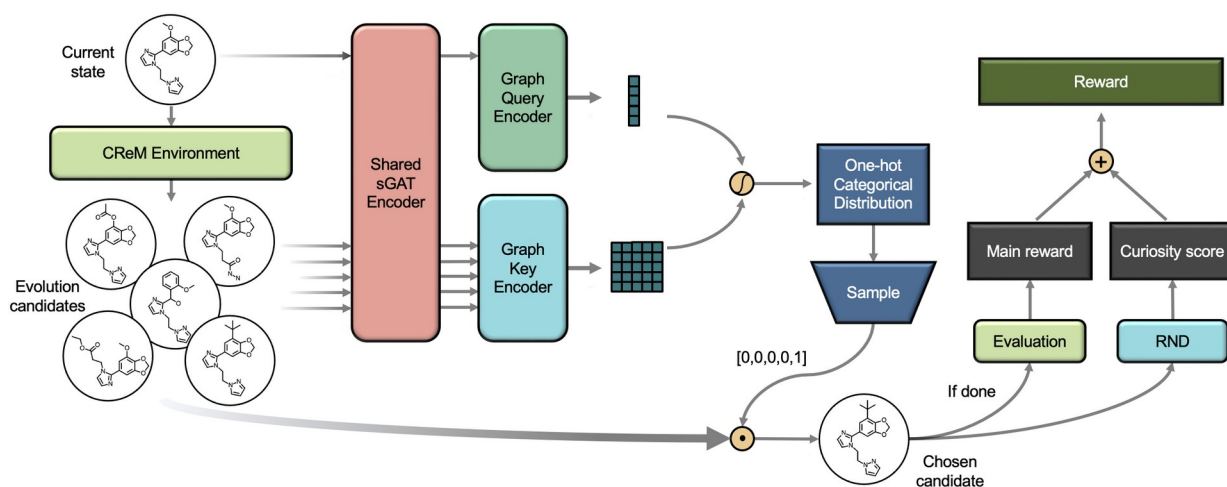


Figure 1. One step of DGAPN's generation loop. The current molecule passes through the CReM environment, which proposes candidates reachable by swapping a single fragment. A shared sGAT encoder produces Query and Key representations; an attention score over the candidates yields a categorical distribution that is sampled

to pick the next molecule. The reward sums a main reward (Evaluation, e.g. the docking score) and a curiosity score (RND).

Paper: <https://arxiv.org/abs/2106.02190>

Code: <https://github.com/yulun-rayn/DGAPN>

The problem: both representation and exploration are hard

Manual drug discovery is slow and expensive, which is exactly what AI-driven molecular design is meant to relieve. But before an algorithm can navigate chemical space on its own, it has to answer two basic questions. How do you compress a molecule into a feature representation a model can learn from, without throwing away the information that matters? And how do you explore a discrete, enormous space efficiently instead of circling in place? Spatial structure matters here because a molecule's volume and the geometry of its interaction sites govern how well it complements a receptor's binding pocket — which is what antiviral efficacy comes down to.

The method: generation as a Markov decision process

DGAPN casts molecule generation as a Markov decision process (MDP). At each step, the CReM chemical library proposes a set of valid molecules reachable by swapping a single chemical fragment — which guarantees, by construction, that every candidate is chemically valid and synthesizable. The policy network then selects one of those candidates as the next molecule. Because an action is simply the choice of the next state, transitions are deterministic; the trajectory length is also fixed and kept short, which stabilizes training and makes it easy to run docking scores in parallel.

Spatial graph attention: a representation that reads both chemistry and geometry

sGAT is the representational core. Its attention is node-centric: when an atom updates its embedding, it draws on information from neighboring atoms and neighboring bonds at once (the different colors of dashed lines in Figure 2 denote different attention weights), and bonds are embedded from the atoms at their endpoints. In this way, the long-neglected bond attributes are finally put to use.

The spatial pathway is what sets sGAT apart. In parallel, it encodes 3D structure from a sparsified inverse distance matrix: the closer two atoms are, the larger the weight, while connections that are too distant are simply dropped (the red crosses in Figure 2 mark the omitted weights). The chemical-attribute pathway and the spatial pathway each yield their own hidden representation, and the two are aggregated at the end of every layer — so the model knows both what a molecule is made of and what it looks like.

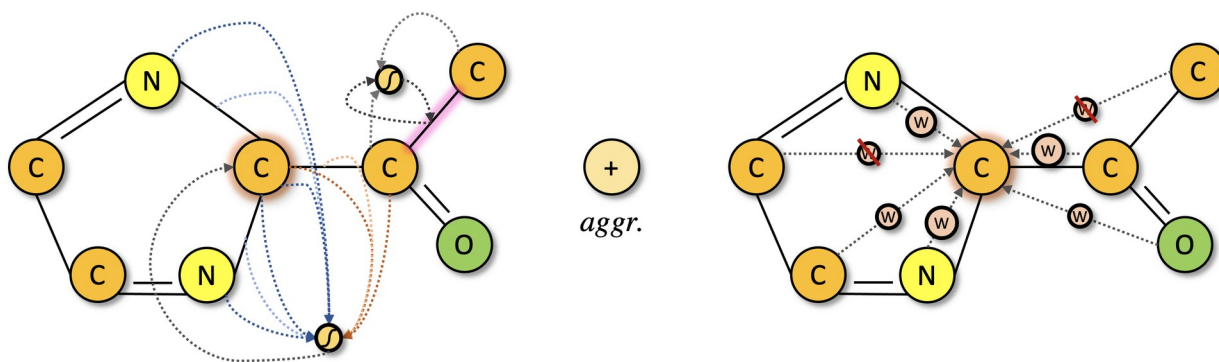


Figure 2. Overview of spatial Graph Attention (sGAT). Left: node-centric attention, where the highlighted atoms and bond are the example undergoing forward propagation and different colors of dashed lines denote different attention weights. Right: spatial structure encoded from a sparsified inverse distance matrix, with red crosses marking omitted weights. The two hidden representations are aggregated (aggr.) at the end of each layer.

The GAPN policy and the curiosity bonus: knowing how to choose, and daring to explore

With a good representation in hand, GAPN (the Graph Attention Policy Network) makes the decisions. It uses attention to measure how well each candidate matches the current state, turns those scores into a categorical distribution over candidates, and samples from it. The policy is trained with proximal policy optimization (PPO) in an actor-critic setup, traded for stability.

Knowing how to choose is not enough; the model also has to dare to explore. The authors adapt Random Network Distillation (RND) to graphs: a predictor network is trained to match a fixed, randomly initialized target network, and the larger the prediction error on a given state, the more novel that state is. Clipped and normalized, that error becomes an "innovation reward" that pushes the agent toward molecules it has not seen. The final reward is the main reward (for example, the docking score) plus this curiosity score.

Experiments: designing inhibitors for NSP15

The main experiment targets the design of novel inhibitors for the functional site of the SARS-CoV-2 NSP15 protein. Binding affinity is estimated by molecular docking, computed with an automated docking tool that runs on GPU from the protein's 3D structure — a lower (more negative) docking score means stronger binding. The authors start from a set of purchasable molecules (Kiss et al., 2012) together with their docking scores against NSP15; molecules that fail conformer generation are scored 0 under an "adjusted validity" criterion. To connect with the broader ML molecule-generation literature, the method is also evaluated on two standard tasks: optimizing the drug-likeness score QED and penalized LogP.

Results: a large lead on docking, with more synthesizable molecules

On the single-objective docking task against NSP15, DGAPN clearly outperforms all five state-of-the-art baselines (REINVENT, JTVAE, GCPN, MolDQN, MARS). It reaches a best docking score of -10.07 and a mean of -6.77 , both well ahead of the runner-up, MolDQN. Under Welch's t-test, the p-value against that second-best model is 8.55×10^{-209} — small

enough that the gap is all but impossible to attribute to chance. Notably, as a fragment-based method DGAPN produces molecules with a synthetic accessibility (SA, lower is better) of 2.91, markedly better than the atom-by-atom MolDQN at 5.04.

Table 1. Primary and summary metrics under single-objective optimization for NSP15 (more negative docking is better; lower SA is better).

Model	Top docking	Mean docking	QED	SA
REINVENT	-8.38	-4.66	0.88	7.51
JTVAE	-7.48	-4.55	0.87	2.65
GCPN	-6.34	-3.18	0.91	5.48
MolDQN	-8.01	-5.38	0.88	5.04
MARS	-8.11	-5.11	0.89	3.38
DGAPN	-10.07	-6.77	0.81	2.91

Under similarity-constrained optimization, DGAPN gives the largest property improvements at every similarity threshold, ahead of both MolDQN and MARS. And in multi-objective mode, a single weight ω trades binding affinity against drug-likeness: at $\omega = 0.6$, DGAPN's molecules reach QED 0.72 and SA as low as 2.20 while still keeping a mean docking score of -5.69 — better on docking than any single-objective baseline.

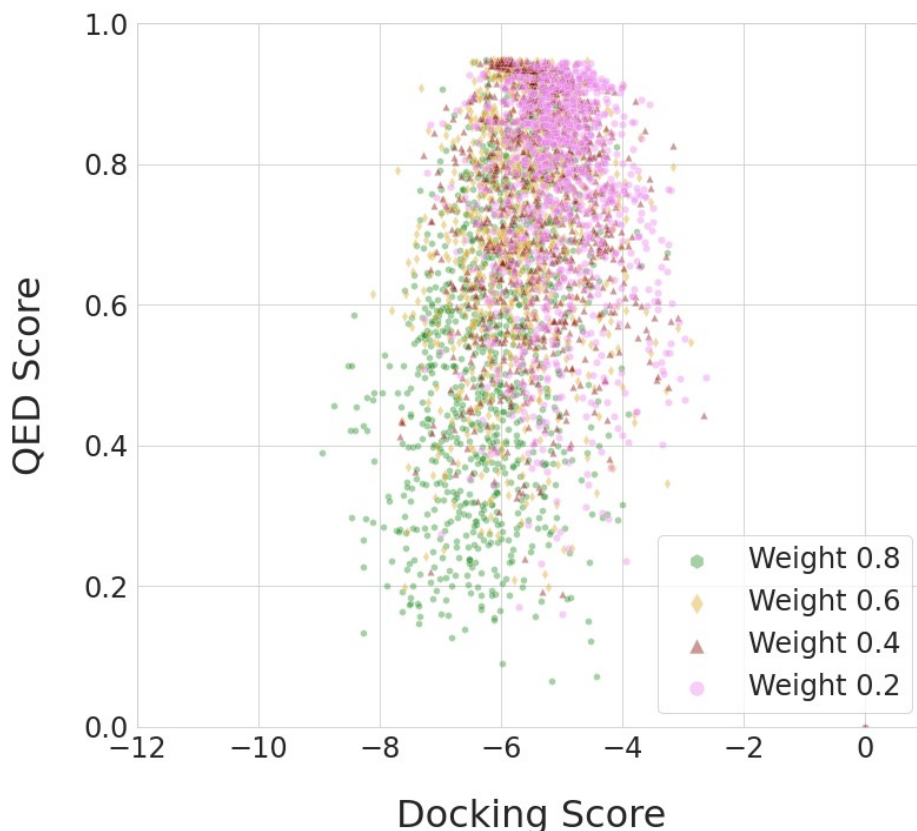


Figure 3. QED versus docking score for the latest 1,000 molecules generated in multi-objective mode, colored by weight ω . As the weight leans toward QED (e.g. 0.2), molecules cluster in the high-QED region; as it leans toward docking (e.g. 0.8), they stretch toward stronger binding (more negative docking scores).

Ablations: every piece is pulling its weight

The authors take the model apart to see what each component buys. First, isolating spatial graph attention in a purely supervised representation-learning setting, the loss curves over 40 runs on the NSP15 data show that the spatial convolution has a clear positive effect on learning molecular representations. Second, comparing the full DGAPN against GAPN without the innovation reward, the curiosity bonus lifts the best docking score from -9.19 to -10.07 — at the cost of slightly worse synthetic accessibility, exposing a docking-versus-SA trade-off. Third, DGAPN even beats a greedy CReM oracle that can see the docking reward of every intermediate candidate, which suggests that the environment design and the stochasticity of the action space are themselves powerful drivers of exploration.

Takeaway: spatial awareness plus curiosity is a good combination for molecular design

DGAPN's core claim is a simple one: encode a molecule's 3D geometry through spatial graph attention, act over whole chemical fragments for validity and synthesizability, and reward novelty with random network distillation — and the model will generate candidates that bind more tightly and are easier to make than those from prior state-of-the-art generators. Just as usefully, the objective is user-defined, so a practitioner can dial the knob between pure binding affinity and a more balanced, drug-like result. The authors are also honest about the boundary: the exploration that curiosity buys comes at a small cost in synthetic accessibility, and how to strike that balance still depends on the design goal at hand.